

Adam Turaj

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EDUCATION

University of Waterloo

Waterloo, ON

BASc — Mechatronics Engineering (Co-op) **GPA: 3.9/4.0**

September 2025 – April 2030 (Expected)

TECHNICAL SKILLS

CAD / Drawings: SolidWorks, Fusion 360, AutoCAD, technical drawings, GD&T

Manufacturing: 3D printing, DFM, DFA, tolerance analysis, composite layup & infusion

Programming: Python, C++, SystemVerilog, Bash

PROJECTS

EOS3D — Budget Cantilever 3D Printer 🔄

February 2025 – August 2025

- Designed and built a cantilever-style 3D printer within a **\$300 budget** with a build volume of **180×180×180 mm**.
- Developed a V2 redesign to improve rigidity, part cooling, and electronics layout.
- Used **Fusion 360**, CFD, and DFM to guide design changes for improved manufacturability and print quality.

Studprint — LEGO-Based Bed-Slinger 3D Printer 🔄

February 2026 – April 2026

- Engineered a **75×64×48 mm** LEGO-based bed-slinger 3D printer in **SolidWorks** using **92% LEGO parts** by count.
- Designed custom 3D-printed mounts for the motors, hotend, and electronics, and used a Bowden extruder to reduce moving mass.
- Iterated the worm gear motion and extrusion system to improve reliability, successfully printing a full-size Benchy on the first attempt.

Skewbot — Skewb Solving Robot 🔄

October 2025 – December 2025

- Co-designed a **4-motor** Skewb-solving robot in **SolidWorks** using VEX hardware and six custom 3D-printed parts.
- Tested prototype performance, achieving a fastest solve time of **10s** and **80% completion reliability**.
- Redesigned grippers and motor mounts through troubleshooting, improving reliability by **60%** while eliminating repeat failures.

EXPERIENCE

Formula SAE (Student Team)

University of Waterloo

Aerodynamics / Electrical Team Member

September 2025 – Present

- Designed a compact **4-layer** high-voltage battery breakout PCB, integrating sensor interfaces, control lines, and **15A** fan power distribution.
- Consolidated **15 discrete cable bundles** into a single connectorized system with integrated PTC fusing, ISO-SPI isolation, and I2C extension.
- Developed a **915 MHz** radio board targeting **2 km range** with integrated SAI and audio codec for combined driver headset and external speaker output.
- Prototyped a coreless composite airfoil process using dissolvable **3D-printed PVA cores** and Mylar-wrapped molds, achieving a **60% weight reduction** versus foam-core designs.

Ciena

Ottawa, Canada

FPGA Verification Co-op

January 2026 – April 2026

- Built and deployed a verification testbench for a production core used across **4 projects**.
- Resolved an intermittent **1-in-16 regression failure** through root-cause analysis and validation with a senior engineer.